

Literature-implied answer for arXiv:0711.3768

Removing the infinite-dimensional Hilbert restriction in the convex-transitive $C_0(L, H)$ theorem

Status

This is a lightweight literature-implied-answer packet. It records that a later theorem answers a source-paper question after direct identification. It is not a new proof packet.

Original Question

In Section 2 of Talponen's *Convex-transitivity and function spaces* [1], in the subsection on $C_0(L, X)$, the paper proves the following two-part theorem:

1. if X is strictly convex and $C_0(L, X)$ is convex-transitive, then X is convex-transitive;
2. if H is a Hilbert space, $\dim(H) = \infty$, and scalar $C_0(L)$ is convex-transitive, then $C_0(L, H)$ is convex-transitive.

Immediately after the theorem, the source says that the proof relies strongly on $\dim(H) = \infty$, and that it is not known whether this assumption can be removed.

Supporting Theorem

Section 4, Theorem 4.3 of Rambla-Barreno and Talponen's *Uniformly convex-transitive function spaces* [2] states: if $C_0^{\mathbb{R}}(L)$ is convex-transitive and X is a convex-transitive Banach space over $\mathbb{K}$, then $C_0^{\mathbb{K}}(L, X)$ is convex-transitive.

Identification

Take $X = H$. Every Hilbert space is convex-transitive: for any two unit vectors, a surjective linear isometry sends one to the other, so every unit vector is a big point. Thus Theorem 4.3 of [2] applies to every Hilbert space H , including finite-dimensional H . The case $\dim(H) = 1$ is the scalar $C_0(L)$ case itself.

Therefore the infinite-dimensional Hilbert hypothesis in the source theorem is not needed. The relation is literature-implied rather than explicitly announced: the 2009 theorem is broader and does not appear to single out the exact 2007 sentence.

Scope

This packet answers only the source-paper question about removing $\dim(H) = \infty$ from the Hilbert-valued $C_0(L, H)$ convex-transitivity theorem. It does not address unrelated open questions in the same source, such as the separate question about $L^\infty(X)$ and convex-transitivity.

References

- [1] J. Talponen, *Convex-transitivity and function spaces*, arXiv:0711.3768v2.
- [2] F. Rambla-Barreno and J. Talponen, *Uniformly convex-transitive function spaces*, arXiv:0902.0640v2.